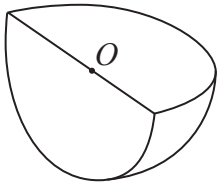
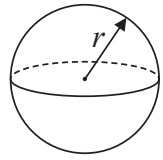
19 Shape **S** is one quarter of a solid sphere, centre O.



Shape **S**

Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



The volume of **S** is 576π cm^3

Find the surface area of **S**.
Give your answer correct to 3 significant figures.
You must show your working.

$\frac{1}{4} \times \frac{4}{3} \quad \pi r^3 = 576\pi$

1 mark
Use one quarter of the sphere volume

Cancel the 4s, then multiply both sides by 3:

$\pi r^3 = 1728\pi$

$\div \pi \qquad \div \pi$

$r^3 = 1728$

cube root cube root

$r = 12$ cm

1 mark
Find the radius

Curved area $= \frac{1}{4} \times 4\pi \times 12^2 = 144\pi$ cm^2

1 mark
Find a surface area component

Two flat semicircles: $2 \times \frac{1}{2}\pi \times 12^2 = 144\pi$ cm^2

Total $= 144\pi + 144\pi = 288\pi$ cm^2

1 mark
Include both flat faces and the curved face

$288\pi = 904.778684...$ cm^2

1 mark
Give three significant figures

905 cm^2

(Total for Question 19 is 5 marks)